

Finmag Porting Decisions — Owner Meeting Checklist

Meeting date: 2026-07-23

Repository owner: Hans Fangohr

Others present: Sam Holt

For each row, tick exactly one:

- **Now** — needed for the first usable serial simulator release (SR1)
- **Later** — required eventually, but can follow SR1
- **Not now** — not required for SR1; retain it in the backlog and reconsider later. This does not mean “never.”

The status column does not use “partial” or “mostly.” It says exactly what is working and what is missing or broken. “Working and tested” applies only to the scope named in that row; it does not imply that the whole subsystem is complete.

Write a short note where the scope is unclear. A permanent omission requires a separate explicit owner decision in `acceptance-register.md`; a **Not now** tick does not authorise deletion or removal from the parity backlog.

1. Core simulator and everyday physics

Functionality	Exact current boundary	Now	Later	Not now	Notes
Pixi install, native build and core import finmag	Working and tested. The editable install, native build, provenance check and core import pass CI. Some optional top-level names still trigger raw dolfin imports; see their separate row.	[x]	[]	[]	
Legacy Simulation constructor, state and common methods	Working and tested: construction, set_m, state access, interactions, energy, run_until and relax. Missing/different: complete public-surface audit, callable pin masks and several convenience methods.	[x]	[]	[]	
Everyday top-level compatibility names	Working: Simulation, sim_with, Field, energies, versions and configuration. Broken: set_logging_level and example can fail through raw legacy-dolfin imports.	[x]	[]	[]	Include finmag.example and barmini in SR1.
Normal-mode top-level compatibility names	Broken: NormalModeSimulation and normal_mode_simulation can fail through raw legacy-dolfin imports. The implementation itself is deferred below.	[]	[x]	[]	
sim_with for common interactions and FK demag	Working and tested: Exchange, Zeeman, uniaxial anisotropy, DMI and FK demag arguments.	[x]	[]	[]	All examples which can me run and all tests needed now. Port all tests and skip/xfail ones we know should not work for now
sim_with for MacroGeometry demag	Not working: direct dense-FK MacroGeometry works, but the factory rejects its nx/ny/spacing arguments.	[x]	[]	[]	MacroGeometry is needed now.
sim_with Treecode selector	Not working: direct Treecode construction works, but the factory rejects a non-FK solver choice.	[]	[x]	[]	Reconsider later; it may not be needed.
Common mesh generators	Working and tested: box, sphere, cylinder, ellipsoid, disks, ring, cone, common CSG combinations and the example .geo subset.	[x]	[]	[]	
Common Field operations	Working and tested: constants, callables, arrays, set/get, ordering, averages, normalisation, point probing, spherical conversion and VTK/XDMF write.	[x]	[]	[]	
Remaining Field operations	Not implemented: from_generic_vector, cross, dot, scalar coercion, HDF5 and plotting.	[x]	[]	[]	
Deterministic serial LLG dynamics	Working and tested: composed effective fields, deterministic trajectories, indexed pins, run_until and relaxation.	[x]	[]	[]	General MPI stepping is a separate Later row.
Callable pin masks	Not implemented: indexed pins work, but coordinate-based Python callables are not accepted.	[x]	[]	[]	
Exchange with supported box assembly	Working and tested: constant and spatially varying A.	[x]	[]	[]	The obsolete matrix/project/direct algorithms are separate Not-now rows.
Static applied fields	Working and tested: Zeeman and DipolarField values, fields and energies.	[x]	[]	[]	
Time-dependent applied fields	Working and tested: TimeZeeman, DiscreteTimeZeeman, TimeZeemanPython scalar mode and OscillatingZeeman. Missing/different: vector-valued TimeZeemanPython and the stale-energy decision below.	[x]	[]	[]	
Uniaxial anisotropy	Working and tested: constant/varying coefficients and axes.	[x]	[]	[]	

Functionality	Exact current boundary	Now	Later	Not now	Notes
Cubic anisotropy	Working and tested: constant axes and constant/varying K coefficients through both field paths. Not implemented: spatially varying cubic axes.	[x]	[]	[]	
DMI documented variants	Working and tested: bulk/auto, interfacial and documented 1D/2D forms, including varying D.	[x]	[]	[]	
DMI D2D variant	Not implemented: raises a D2D-specific error.	[]	[x]	[]	
FK demagnetisation	Working and tested: compiled FK field and energy against frozen-oracle references.	[x]	[]	[]	
MacroGeometry demagnetisation	Working and tested when constructed directly: dense-FK analytic/cross-method checks. Missing: <code>sim_with</code> wiring is the separate Now row above.	[x]	[]	[]	
Treecode demagnetisation	Working and tested when constructed directly. Missing: <code>sim_with</code> selector is the separate Later row above.	[]	[x]	[]	Reconsider whether this is required.
Spatially varying materials and regions	Working and tested: varying Ms, A, K, D, axis and alpha; region marking, energies and averages. Not implemented: region-restricted field output.	[x]	[]	[]	
Slonczewski STT	Working and tested when used alone: oracle/analytic and both-integrator checks. Different: simultaneous behavior with Zhang-Li is an owner choice below.	[x]	[]	[]	simultaneous not needed
Zhang-Li STT	Working and tested when used alone: oracle/analytic, domain-wall and both-integrator checks. Different: simultaneous behavior with Slonczewski is an owner choice below.	[x]	[]	[]	simultaneous not needed
Relaxation	Working and tested.	[x]	[]	[]	
Hysteresis	Runs, but is physically wrong after the first stage: the driver changes the applied field but does not perform a fresh relaxation, so later points are not independently equilibrated.	[x]	[]	[]	Fix and document; see Section 4.
Scheduler and NDT output	Working and tested: scheduled actions plus NDT write/read.	[x]	[]	[]	
VTK/XDMF field output	Working and tested for writing. Not implemented: general readback.	[x]	[]	[]	
.npy field snapshots	Working and tested: coordinate/value table format. Different: original files used raw backend dofs; owner choice below.	[]	[x]	[]	
Restart on SciPy	Working and tested for restoring m/time. Missing evidence: restarted trajectory is not compared with an uninterrupted control. Old raw-dof v1 files are rejected.	[x]	[]	[]	
Restart on Sundials	Broken: restart calls reset and fails on a SciPy-only .ode assumption. Saved driver metadata is also hard-coded to <code>scipy</code> .	[x]	[]	[]	Sundials should be default
SciPy integrator lifecycle	Working and tested: construct, advance, tolerance update, reset/reinit and SciPy restart.	[x]	[]	[]	
Sundials/CVODE integration	Working and tested: construction, trajectory advance and SciPy cross-check. Broken: reset and restart.	[x]	[]	[]	
Fast converted examples	Working and tested: 14 pass; 3 full-resolution cases are deliberately skipped.	[x]	[]	[]	

Functionality	Exact current boundary	Now	Later	Not now	Notes
Full-resolution examples and muMAG standard problem 4	Not yet accepted: no recorded all-green full lane. The current std_prob4 0.10–0.18 ns switching window is qualitative, not proof of unchanged behavior.	[x]	[]	[]	We need to make sure all the tests are ported as close to they previously were in master/pixi this will highlight any changes in behaviour or interfaces. Including standard problems

2. Specialist physics and analysis

Functionality	Exact current boundary	Now	Later	Not now	Notes
Thermal stochastic LLG (SLLG)	Not implemented on DOLFINx.	[]	[]	[x]	
Landau-Lifshitz-Bloch (LLB) and material laws	Not implemented on DOLFINx.	[]	[]	[x]	
Linear normal modes and eigensolvers	Not implemented on DOLFINx; top-level access can reach raw dolfin imports.	[]	[x]	[]	
Ringdown analysis, FFT/PSD and dispersion	Not implemented on DOLFINx.	[]	[x]	[]	
Legacy NEB variants	Not implemented on DOLFINx. All master NEB variants still need an algorithm/API inventory.	[]	[x]	[]	
GNEB as a named workflow	No public GNEB API/workflow found on master. Possible overlap with spherical/modified legacy NEB has not been classified.	[]	[x]	[]	To be clear one GNEB method would be good but we will do this in a later PR
Nonlocal LLG_STT spin-accumulation model	Not implemented on DOLFINx. This is distinct from the two working local STT terms.	[]	[]	[x]	
General MPI-parallel time stepping	Not implemented. Only ownership-sensitive Field/energy probes currently run on multiple ranks.	[]	[x]	[]	
Simulation(pbc=) function-space periodic boundaries	Not implemented. Treecode periodic MacroGeometry is a different, working capability.	[x]	[]	[]	
Demag2D	Not implemented on DOLFINx.	[]	[]	[x]	
GCR demag	Not implemented. The surviving legacy Python formulation appears incomplete and its intended scientific contract needs review.	[]	[]	[x]	
FixedEnergyDW	Not implemented. No master tests were found and a legacy note calls the class broken.	[]	[]	[x]	

3. I/O, utilities and external tools

Functionality	Exact current boundary	Now	Later	Not now	Notes
HDF5 readback	Not implemented on DOLFINx.	[x]	[]	[]	
Plotting and visualisation helpers	Not implemented on the supported DOLFINx path. VTK/XDMF data writing is covered separately above.	[x]	[]	[]	
Region/submesh field output	Not implemented. Region energies and averages do work.	[x]	[]	[]	
Magnetisation initialisers (<code>initialise_*</code>)	Not implemented on DOLFINx.	[x]	[]	[]	
<code>length_scales</code> , <code>mesh_info</code> , profiling, logging/instance/shutdown helpers	Not implemented or still tied to legacy imports. These names require individual inventory before any omission decision.	[x]	[]	[]	
<code>Simulation.probe_field*</code> , LLG M/M_average and remaining Field arithmetic	Not implemented on the supported DOLFINx surface.	[x]	[]	[]	
Specialist mesh generators	Not implemented: layered/internal-shell/polygon/line/embed3d and other long-tail generators. Common generators are covered above.	[]	[]	[x]	
Netgen binary backend	Not implemented. Gmsh and a limited textual .geo parser cover the current common subset.	[x]	[]	[]	
<code>nmesh_to_dolfin</code> conversion	Not implemented; its output is legacy dolfin-XML, which DOLFINx cannot read.	[]	[]	[x]	
OOMMF comparison workflow	Live harness not ported. Some ported tests use checked-in cross-code/reference data.	[x]	[]	[]	
Nmag/nsim comparison workflow	Live harness not ported; <code>nsim</code> is absent. Some checked-in Nmag reference data is exercised.	[x]	[]	[]	
Magpar comparison workflow	Live harness not ported. Saved data exists, but some historical comparisons have mesh/node-order drift.	[x]	[]	[]	
Legacy matrix energy assembly	Not implemented; box assembly is the supported algorithm.	[]	[]	[x]	
Legacy project energy method	Not implemented; box assembly is the supported algorithm.	[]	[]	[x]	
Legacy direct energy method	Not implemented; box assembly is the supported algorithm.	[]	[]	[x]	
Varying cubic-anisotropy axes	Not implemented; constant axes work.	[x]	[]	[]	
Legacy string Expression compatibility	Not implemented as strings. Old FEniCS accepted formula text that it compiled at runtime; DOLFINx users can supply ordinary vectorised Python callables instead.	[]	[]	[x]	Do not add string compatibility; document callable examples and clear errors.
Compiled Equation/terms performance backend	Not implemented. A slower Python fallback exists; DOLFINx LLG uses its own RHS.	[]	[x]	[]	How important is this?
Paraview/mencoder movie wrapper	Not implemented. VTK/XDMF simulation data can be written for external tools.	[]	[x]	[]	Can we use pyvista instead?
<code>batch_task</code> sweep tooling	Not implemented and untested on master.	[]	[]	[x]	
Mercurial revision helper	Not ported; obsolete in the Git repository.	[]	[]	[x]	

4. Behavior decisions

Decisions discussed with Hans Fangohr and relayed on 2026-07-23:

Topic	Decision and implementation contract
Stale <code>DiscreteTimeZeeman.compute_energy()</code>	Fix and document. The field and reported energy must both update; retain a regression explaining the legacy bug.
Hysteresis no-re-relax defect	Fix and clearly document. Every field stage must independently relax; retain a regression explaining why legacy output differed.
Default integrator	Sundials. Restore the legacy public default only after reset, reinitialisation, scheduling and restart pass the same lifecycle checks as SciPy.
Simultaneous Slonczewski and Zhang-Li STT	Reject explicitly. Each mode works separately; attempting to configure both must raise a clear error rather than silently applying precedence or last-call-wins.
Invalid pin index	Raise ValueError. Do not preserve legacy behavior that logged and silently retained stale pins.
Restart format	Version 2 only. There are no legacy restart files to support; reject v1 clearly rather than attempting unsafe raw-dof remapping.
Field snapshot format	No raw-format compatibility. Retain the coordinate/value representation; do not reproduce backend-dof arrays.
Spatially varying cubic K2 indexing	Use correct physics. Do not reproduce the legacy native indexing bug.
Legacy string Expression inputs	No backward compatibility. Use documented vectorised Python callables and clear errors for string inputs.

Meeting outcome

SR1 must contain:

The rows ticked **Now** are the canonical SR1 scope. In particular: deterministic serial dynamics; callable pins; MacroGeometry-through-FK; PBC probe and serial implementation; selected Field/I/O/convenience rows; checked reference-data comparisons; and the legacy-test translation inventory.

Definitely later:

The rows ticked **Later** are the canonical later backlog, including normal modes, ringdown/FFT, NEB/GNEB work, MPI stepping, Treecode factory exposure, and movie tooling.

Not required in the short term; retain for later review:

The rows ticked **Not now** are retained, not dropped: thermal SLLG/LLB, nonlocal STT, Demag2D/GCR/FixedEnergyDW, specialist mesh/tooling paths and legacy energy assembly. Section 4's per-behaviour decisions are canonical for the meeting record; the acceptance register is canonical for approved corrections and compatibility changes.

P0.2 baseline:

P0.2 baseline evidence was recorded on 2026-07-23.

Post-meeting implementation status (2026-07-23)

The ticks and status text above are the dated owner-meeting snapshot; they are not silently rewritten as implementation lands. Since that meeting, P1.1 and P1.2 fixed the three meeting-time lifecycle/restart boundaries: backend-neutral reset/reinitialisation, Sundials restart, and truthful restart-driver provenance. The following dated addendum records the completed P1.3 result; see `capability-status.md` and `acceptance-register.md` for the canonical current status and D8 disposition.

Post-meeting implementation addendum (2026-07-23)

P1.3 is complete in 81fab481. It restores native Sundials as the public default for `Simulation` and `sim_with`, discharging D8's conditional approval. The actual native `SundialsIntegrator` is constructed and advanced by the default path; the core smoke records `"integrator_backend": "sundials"` at `t=1e-12`. Explicit `integrator_backend="scipy"` remains fully supported.

Focused P1.3 evidence: `Simulation` 37 passed, `Sundials` 22 passed, `SciPy` 21 passed/2 skipped, `restart/output` 27 passed. The final clean main-worktree aggregate on 81fab481 had all 32 steps green; fast examples were 14 passed/3 skipped in 246.36s (`/tmp/finmag-p1-default-m5.log`). This does not alter the meeting snapshot or make the frozen P0.2 FULL lane green: that lane remains blocked at 12 passed/5 failed.

[Codex GPT-5]